

1 Flow Chart

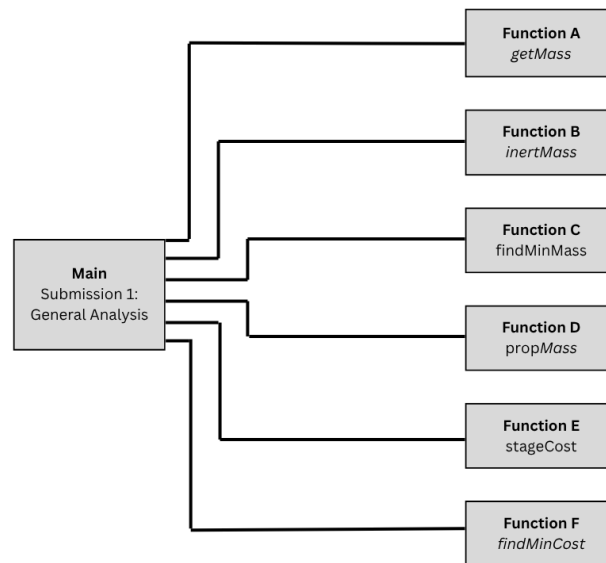


Figure 1: Flow Chart of Function Organization

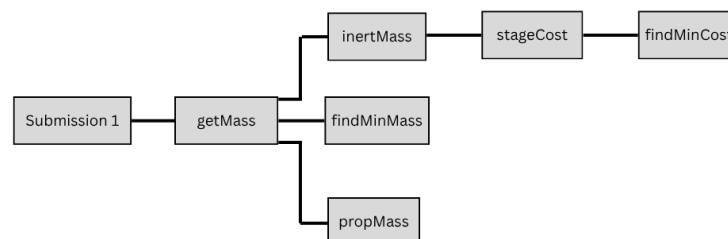


Figure 2: Flow Chart of Function Call Order

2 Code

```
1 % Submission 1 Template, Group 1
2
3 close all;
4 clear;
5 clc;
6 % Givens
7 delta_v = 12.3; % km/s
8 m_pl = 26000; % kg
9 delta = 0.08; % Inert mass fraction for both stages
10 chi = 0.2:0.01:0.8; %array
11 Isp1 = 327; % s, 1st stage, LOX/CH4
12 %      LOX/CH4 LOX/LH2 LOX/RP1 Solid   Storables
13 Isp2 = [327,    366,    311,    269,    285]; % s, 2nd stage
14 stage1Prop = "LOX/CH4";
15 stage2Prop = ["LOX/CH4", "LOX/LH2", "LOX/RP1", "Solid", "Storables"];
16
17
18 for k=1:length(Isp2)
19     [M01_array, M02_array, chi_array] = getMass(delta_v, m_pl, delta, chi, Isp1,
20         Isp2(k));
21     [m_pr1, m_pr2] = propMass(delta, M01_array, M02_array, m_pl);
22     [m_in1, m_in2] = inertMass(delta, M01_array, M02_array);
23     Mo = M01_array + M02_array;
24
25     M01_array = M01_array./1000;
26     M02_array = M02_array./1000;
27     Mo = Mo./1000;
28
29     [minMass, mIndex] = findMinMass(Mo);
30     minMass
31     costForMinMassS1 = stageCost(m_in1(mIndex));
32     costForMinMassS2 = stageCost(m_in2(mIndex));
33     costForMinMass = costForMinMassS1 + costForMinMassS2
34     costS1 = stageCost(m_in1);
35     costS2 = stageCost(m_in2);
36     costTotal = costS1 + costS2;
37     [minCost, cIndex] = findMinCost(costTotal);
38     minCost
39     massForMinCost = Mo(cIndex)
40
41     l = round(length(chi_array)/4);
42     figure(k)
43     hold on;
44     grid on;
45     plot(chi_array(mIndex-l:mIndex+l), M02_array(mIndex-l:mIndex+l));
46     plot(chi_array(mIndex-l:mIndex+l), M01_array(mIndex-l:mIndex+l));
47     plot(chi_array(mIndex-l:mIndex+l), Mo(mIndex-l:mIndex+l));
48     plot(chi_array(mIndex), minMass, "o");
49     legend('M_{02}', 'M_{01}', 'M_0', "M_{min}");
50     titleMass = sprintf("Total Masses vs. Chi (S1: %s, S2: %s)", stage1Prop,
51         stage2Prop(k));
52     title(titleMass);
```

```
51     ylabel('Mass (tonnes)');
52     xlabel('Chi');
53
54     figure(length(Isp2) + k)
55     hold on;
56     grid on;
57     plot(chi_array(cIndex-1:cIndex+1), costS2(cIndex-1:cIndex+1));
58     plot(chi_array(cIndex-1:cIndex+1), costS1(cIndex-1:cIndex+1));
59     plot(chi_array(cIndex-1:cIndex+1), costTotal(cIndex-1:cIndex+1));
60     plot(chi_array(cIndex), minCost, "o");
61     legend('Cost_2', 'Cost_1', 'Cost_0', "Cost_{min}");
62     titleMass = sprintf("Cost vs. Chi (S1: %s, S2: %s)", stage1Prop,
        stage2Prop(k));
63     title(titleMass);
64     ylabel('Cost ($M)');
65     xlabel('Chi');
66
67 end

1 function [M01_array, M02_array, chi_array] = getMass(delta_v, m_pl, delta, chi,
    Isp1, Isp2)
2 % GETMASS Get total stage masses, M01 and M02.
3 %
4 % [M01_array, M02_array, chi_array] = GETMASS(delta_v, m_pl, delta, chi,
    Isp1, Isp2)
5 % Use the rocket equation to determine total stage masses given:
6 % - delta_v: the required change in velocity
7 % - m_pl: the required payload mass
8 % - delta: the estimated inert mass fraction
9 % - chi: the target delta V split between the first and second stage
10 % - Isp1: the specific impulse of the first stage engines
11 % - Isp2: the specific impulse of the second stage engines
12 %
13 % See also PROPMASS, INERTMASS, FINDMINMASS.
14 syms M02 M01
15 %Initialize mass arrays
16 M02_array = [];
17 M01_array = [];
18 chi_array = [];
19 g = 9.81/1000;%km/s^2
20
21 for k = 1:length(chi)
22     %Rocket equation for stage two
23     eqn2 = -Isp2*g*log((m_pl+delta*M02) / M02) - (1-chi(k))*delta_v == 0;
24     eqn1 = -Isp1*g*log((M02+delta*(M01+M02)) / (M02+M01)) - chi(k)*delta_v
        == 0;
25     soln = vpasolve([eqn1, eqn2],[M01, M02]); %Use vpa solver to find M02
26
27     %The below loop excludes mass values that are negative and
28     %populates the matrices for both stages
29     if (soln.M02 > 0 && soln.M01 > 0)%
30         try
31             M02_array(end+1) = double(soln.M02); %Store mass values
32             M01_array(end+1) = double(soln.M01); %Store mass values
```

```
33         chi_array(end+1) = [chi(k)]; %Store corresponding chi values
34     catch
35     end
36 end
37 end
38
39 end

1 function [m_in1, m_in2] = inertMass(delta, M01_array, M02_array)
2 % Emma worked on this code
3 % INERTMASS Find the inert mass given the inert mass fraction and the stage
  masses.
4 %
5 % [m_in1, m_in2] = INERTMASS(delta, M01_array, M02_array) Finds the
6 % inert mass of stage 1 (from M01_array) and the inert mass of stage 2
7 % (from M02_array) using inert mass fraction delta.
8 %
9 % See also GETMASS, PROPMASS, FINDMINMASS.
10 m_in1 = delta.*M01_array;
11 m_in2 = delta.*M02_array;
12 end

1 function [m_pr1, m_pr2] = propMass(delta, M01, M02, m_pl)
2 % All team members worked on this code
3 % PROPMASS Find the propellant masses from the total stage masses.
4 %
5 % [m_pr1, m_pr2] = PROPMASS(delta, M01, M02, m_pl) Find propellant masses
6 % for stage 1 and 2 (m_pr1 and m_pr2), from total stage masses for stage 1
  and 2 (M01 and
7 % M02), using inert mass fraction (delta) and payload mass (m_pl).
8 %
9 % See also GETMASS, INERTMASS, FINDMINMASS.
10 m_pr1 = []; % establish array
11 m_pr2 = [];
12 for k = 1:length(M01) % for length of stage 1 total mass array
13     m_in1 = delta*M01(k); % inert mass formula for stage 1
14     m_pr1(k) = M01(k) - m_in1; % mass of propellant stage 1
15     m_in2 = delta*M02(k); % inert mass formula for stage 2
16     m_pr2(k) = M02(k) - m_pl - m_in2; % mass of propellant stage 2
17 end
18 end

1 function [mass, index] = findMinMass(Mos)
2 % Joseph worked on this code
3 % FINDMINMASS Find the minimum mass given a list of total vehicle masses.
4 %
5 % [mass, index] = FINDMINMASS(Mos) Finds the minimum mass in Mos.
6 %
7 % See also FINDMINCOST.
8
9
10 % call min function to find the min total mass, return the mass value
11 % and index
12 [mass, index] = min(Mos);
13 end
```

```
1 function cost = stageCost(stageInertMass)
2 % STAGECOST Find the cost of a stage based on inert mass.
3 %
4 % [cost] = STAGECOST(stageInertMass) Find the cost of a stage in millions
5 % of 2025 dollars based on the stageInertMass
6 %
7 % See also FINDMINCOST, INERTMASS.
8 cost = 13.52.*stageInertMass.^0.55;
9 end

1 function [cost, index] = findMinCost(costs)
2 % Joseph worked on this code
3 % FINDMINCOST Find the minimum cost given a list of total vehicle costs.
4 %
5 % [cost, index] = FINDMINCOST(costs) Finds the minimum cost in costs.
6 %
7 % See also FINDMINMASS, STAGECOST.
8
9
10 [cost, index]=min(costs);
11 end
```